

AN ALTERNATIVE NOTATION FOR PIECEMEAL DISTRIBUTIONS

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It is frequently necessary to evaluate the density of a random variable X restricted to a subregion A of its support. The traditional convention,

$$f_X(x) = g(x), \quad x \in A,$$

detaches the functional form from its domain of validity, obscuring boundary mismatches in multi-step calculations.

Let the total domain of a density $f_X(x)$ be partitioned into regions $\{A_i\}$. Define the *piecemeal branch* $f_{X \in A_i}(x)$ as the **unnormalized** slice of the density restricted to that region. The global distribution is reconstructed by the summation:

$$f_X(x) = \sum_i f_{X \in A_i}(x), \quad \text{where} \quad \sum_i \int_{A_i} f_{X \in A_i}(x) dx = 1.$$

For joint and conditional domains restricted to active regions $y \in B$ and $x \in A$, the notation extends to:

$$f_{Y \in B | X \in A}(y | x).$$

Bayesian inference requires strict structural alignment across the expression:

$$f_{\Theta \in B | X \in A}(\theta | x) = \frac{f_{\Theta \in B}(\theta) \cdot f_{X \in A | \Theta \in B}(x | \theta)}{f_{X \in A}(x)}.$$

The denominator $f_{X \in A}(x)$ is the specific unnormalized marginal branch for region A . A subscript mismatch anywhere in the expression immediately signals a boundary violation.

Example. Let $X | \Theta = \theta \sim \text{Uniform}([\theta, 2\theta])$ and $\Theta \sim \text{Uniform}([0, 1])$, such that $f_{X | \Theta}(x | \theta) = 1/\theta$ and $f_{\Theta}(\theta) = 1$. The joint support projects onto two distinct observation regions for X . Integration over the active θ -slices yields the marginal branches:

$$f_{X \in [0, 1]}(x) = \int_{x/2}^x \frac{d\theta}{\theta} = \ln 2, \quad f_{X \in [1, 2]}(x) = \int_{x/2}^1 \frac{d\theta}{\theta} = \ln \frac{2}{x}.$$

Global validity is verified by the sum of the branch integrals:

$$\int_0^1 \ln 2 dx + \int_1^2 \ln(2/x) dx = 1.$$

For an observation $x \in [0, 1]$, the parameter support is constrained to $\theta \in [x/2, x]$. The posterior branch is determined by:

$$f_{\Theta \in [x/2, x] | X \in [0, 1]}(\theta | x) = \frac{1 \cdot (1/\theta)}{f_{X \in [0, 1]}(x)} = \frac{1}{\theta \ln 2}.$$

Substitution of $f_{X \in [1, 2]}$ in the denominator is invalid by inspection of the subscripts. Conditioned on a fixed coordinate x , the posterior branch normalises to unity over the localised θ domain.